

compute a taxable value of money in the GRAT (equation (12.8)). This calculation is more difficult than the simple case because it involves valuing payments in excess of a (pre-determined but stochastic) hurdle rate. Different trusts will have different taxable income conversion factors. Conceptually, Huang's basic model is unchanged.

One disadvantage of trusts is that they can be hard to dismantle. (Many are *irrevocable*.) Of course, not being able to touch your money could be an advantage, but it is a form of liquidity constraint. The standard Merton models, and Huang's tax-adjusted extension of Merton's models, do not take into account illiquidity risk or constraints. In chapter 13, we discuss asset allocation models that do.

4.5. TAXES, CONSUMPTION, INVESTMENT, AND OTHER DECISIONS

Investors also use taxes to their advantage in timing consumption and investment decisions. Some of these can be pure tax arbitrages. A famous example is the pension tax arbitrage of Black (1980) and Tepper (1981). They show that a corporate defined pension fund allows a corporation to engage in a form of tax arbitrage by allowing the corporation (the plan sponsor) to borrow money, take a tax deduction for the interest, and then invest the proceeds in the pension plan. Pension funds are tax exempt and are a natural clientele for highly taxed assets like corporate bonds and U.S. Treasuries. The corporation gets a tax break for the pension fund contribution, and the pension fund gets to house highly taxed assets in a tax-efficient vehicle. Moreover, since the corporation is actually a stock, the Black-Tepper tax arbitrage allows the corporation to harvest an equity risk premium by investing in bonds!

Frank (2002) presents evidence that corporations do undertake the Black-Tepper strategy and shows that there is a positive relation between the amount held by the company's pension plan in bonds and the tax benefit. But the Black-Tepper strategy is vastly inefficiently taken up. Company pension plans seem to hold way too much equity, at least solely according to tax arguments. Perhaps this is optimal according to other arguments, as chapter 1 discusses.

Pension fund contributions are deductible by corporations. Firms strategically time the contributions when tax rates are high. This practice now has limits because contributions when the plan is more than 150% overfunded cannot be deducted under the 1987 Tax Act. The 1990 Tax Act also raised the excise tax to 50% on a corporation taking out assets from an overfunded pension plan. The same concept applies to individuals—deductions for contributions to charities, pensions, and so on are more valuable when tax rates are high.

On a final morbid note, the ultimate economic decision is timing one's death to minimize, or in some cases entirely avoid, income taxes. In the U.S. in 2010, there was no estate tax. Anecdotal evidence suggested that people tried to postpone